

Introduction to Electronics Engineering

PART 2 — QUESTION BANK WITH ANSWERS

Module 1: Semiconductor Diodes and their Applications

Question pattern covered in this PDF:

Section	Marks	Scope	No. of Questions
Section A	2 marks each	Compulsory from every module	16
Section B	5 marks each	Only from Modules 1 and 2	9
Section C	7 marks each	Compulsory from every module	6

Each answer is written the way it should be presented in the examination. 2-mark answers are short and to the point; 5-mark and 7-mark answers are structured in clear points with the relevant diagrams and formulas.

Section A — 2-Mark Questions

Answer each in 2–4 lines. Write the definition and, where asked, one example or one formula.

Q1. What is Electronics? What is an electronic circuit?

Ans. Electronics is the branch of science and engineering that deals with the study of the flow of electrons through vacuum, gas, or semiconductors. An electronic circuit is a closed path formed by the interconnection of electronic components through which electric current flows to perform a specific function.

Q2. What is a semiconductor? Give two examples.

Ans. A semiconductor is a material whose electrical conductivity lies between that of a conductor and an insulator, and whose conductivity can be controlled by adding impurities (doping) or by applying external fields. Examples: Silicon (Si) and Germanium (Ge).

Q3. What are the three classifications of materials based on the forbidden energy gap?

Ans. Conductors have no forbidden gap; semiconductors have a small forbidden gap (about 1 eV); insulators have a large forbidden gap (more than 5 eV).

Q4. What is doping? What are dopants?

Ans. Doping is the deliberate addition of impurities to a pure (intrinsic) semiconductor to alter its electrical properties. The added impurity atoms are called dopants. Doping converts an intrinsic semiconductor into an extrinsic (n-type or p-type) semiconductor.

Q5. Differentiate between n-type and p-type semiconductors.

Ans. An n-type semiconductor is formed by adding pentavalent (5 valence electron) impurities such as antimony, arsenic or phosphorus; the impurity atoms are donors, and electrons are the majority carriers. A p-type semiconductor is formed by adding trivalent (3 valence electron) impurities such as boron, gallium or indium; the impurity atoms are acceptors, and holes are the majority carriers.

Q6. Define conduction band, valence band and Fermi level.

Ans. The conduction band is the energy band where electrons are free to move and contribute to electrical conductivity. The valence band is the energy band where electrons are bound to atoms and do not contribute to conductivity. The Fermi level is the highest energy level an electron can occupy at absolute zero temperature; it lies between the valence and conduction bands.

Q7. What is the depletion region? How does its width depend on doping?

Ans. The depletion region is the region around the PN junction that has no free charge carriers — only immobile ions remain (negative acceptor ions on the p-side, positive donor ions on the n-side). Heavily doped regions have narrower depletion regions.

Q8. What is the barrier (built-in) potential of a PN junction? State its values for Si and Ge.

Ans. The electric field across the depletion region creates a potential barrier that prevents further diffusion of charge carriers across the junction. At room temperature it is 0.7 V for silicon and 0.3 V for germanium.

Q9. What is knee voltage?

Ans. The knee voltage is the minimum forward voltage after which the forward current of a diode starts increasing exponentially. For silicon it is about 0.7 V and for germanium about 0.3 V.

Q10. What is reverse saturation current (leakage current)?

Ans. It is the very small current that flows through a reverse-biased diode, caused by thermally generated minority carriers swept across the junction by the built-in electric field. It is almost constant with reverse voltage and approximately doubles for every 10°C rise in temperature.

Q11. Differentiate between static and dynamic resistance of a diode.

Ans. Static (DC) resistance is defined at one operating point on the I–V curve: $R_D = V_D/I_D$. Dynamic (AC) resistance is found from the slope of the curve at the operating point: $r_d = nV_T/I_D$ (about 26 mV/ I_D for $n = 1$).

Q12. Write Shockley's diode equation and explain the terms.

Ans. $I_D = I_s (e^{V_D/nV_T} - 1)$, where I_D is the diode current, I_s the reverse saturation current, V_D the applied forward voltage, n the ideality factor ($n = 1$ for Ge, $n = 2$ for Si) and $V_T = kT_K/q$ the thermal voltage (about 26 mV at 27°C).

Q13. Distinguish between Zener and avalanche breakdown.

Ans. Zener breakdown occurs in heavily doped diodes with a narrow depletion region, needs a low voltage (typically below 5 V), and has a negative temperature coefficient. Avalanche breakdown occurs in lightly doped diodes with a wide depletion region, needs a high voltage, and has a positive temperature coefficient.

Q14. What are junction capacitance and diffusion capacitance?

Ans. Junction (depletion) capacitance C_j is associated with the depletion region of a reverse-biased diode and decreases as the reverse voltage increases: $C_j = (\text{permittivity} \times \text{area})/\text{width}$. Diffusion capacitance C_d appears in a forward-biased junction due to stored injected charge and increases with forward current.

Q15. Define ripple factor and PIV of a rectifier.

Ans. Ripple factor measures the AC content remaining in the rectifier DC output: it equals the square root of $[(V_{rms}/V_{dc})^2 - 1]$ (1.21 for half-wave, 0.48 for full-wave). PIV (Peak Inverse Voltage) is the maximum reverse voltage a diode must withstand without breaking down.

Q16. Differentiate between a clipper and a clamper.

Ans. A clipper removes (clips) a portion of the input waveform — the shape is changed — and needs only a diode and resistor. A clamper shifts the entire waveform up or down by adding a DC component without changing its shape, and a capacitor is essential for its operation.

Section B — 5-Mark Questions

Q1. Explain the formation of the depletion region in a PN junction (no-bias condition).

When a p-type and an n-type material are joined, electrons from the n-region diffuse into the p-region and holes from the p-region diffuse into the n-region, until equilibrium is established. Near the junction the electrons and holes recombine, creating a **depletion region** — a region with no free charge carriers where only immobile ions remain (negative acceptor ions on the p-side and positive donor ions on the n-side).

The ions set up an electric field across the depletion region, directed from the n-side to the p-side, which opposes further diffusion. When the diffusion is exactly balanced by this field, equilibrium is reached and the net flow of charge in any direction is zero.

- Width depends on doping — heavily doped junctions have narrower depletion regions.
- The field creates a barrier potential V_b : 0.7 V for Si, 0.3 V for Ge at room temperature.

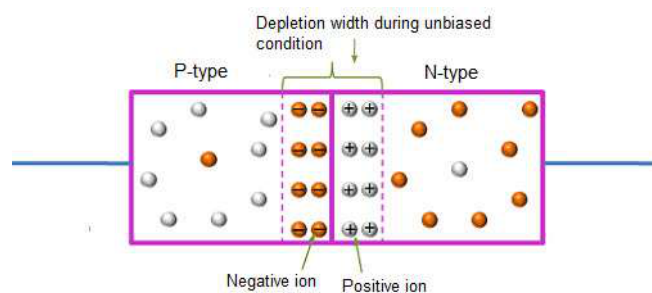


Figure 1: Formation of the depletion region and the barrier potential.

Q2. Explain the forward and reverse bias operation of a PN junction diode.

Forward bias (positive terminal to the p-region): the potential barrier is reduced, carriers cross the junction (electrons toward the p-region, holes toward the n-region), and a significant forward current flows. A silicon diode needs about 0.7 V (germanium about 0.3 V) before it conducts appreciably; beyond this knee voltage the current rises exponentially.

Reverse bias (positive terminal to the n-region): the potential barrier increases and the depletion region widens, so majority carriers cannot cross the junction. Only a very small reverse saturation current (due to minority carriers) flows. If the reverse voltage exceeds the breakdown value, the diode conducts heavily and may be permanently damaged.

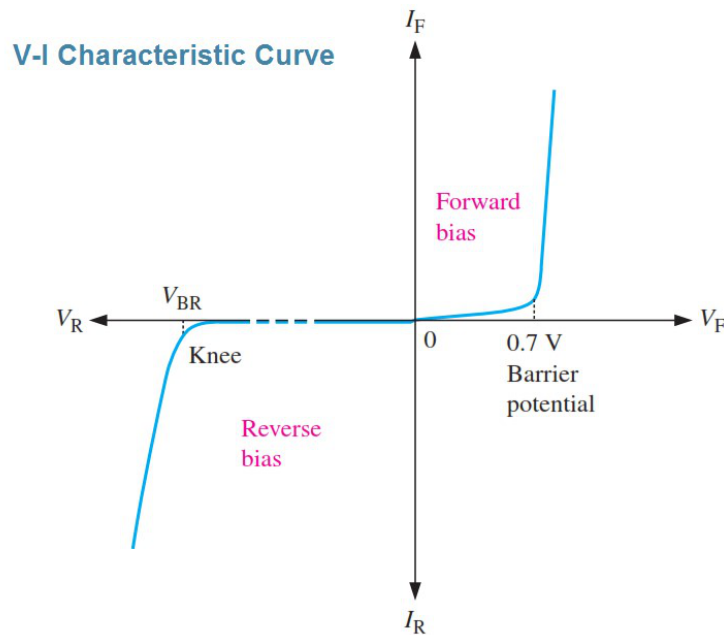


Figure 2: The complete I–V characteristics of a PN junction diode.

Q3. State and explain Shockley's diode equation. What is thermal voltage?

The current–voltage relationship of a semiconductor diode is given by Shockley's equation:

$$I_D = I_s (e^{V_D/nV_T} - 1)$$

- I_D = diode current; I_s = reverse saturation current.
- V_D = applied forward-bias voltage.
- n = ideality factor: $n = 1$ for germanium, $n = 2$ for silicon.
- $V_T = kT_K/q$ = thermal voltage, about 26 mV at 27°C (T in kelvin = $273 + ^\circ\text{C}$).

For a forward voltage greater than a few tens of millivolts the “–1” term becomes negligible and the current rises exponentially with voltage; in reverse bias the current saturates at $-I_s$.

Q4. Explain the effect of temperature on the characteristics of a semiconductor diode.

- Increasing temperature adds energy to the diode and reduces the forward voltage needed for conduction.
- The forward characteristic shifts left (lower voltage) by about 2.5 mV per $^\circ\text{C}$ for silicon.
- The reverse saturation current approximately doubles for every 10°C rise: $I_s = I_{s0} \times 2^{(T-T_0)/10}$.
- Germanium diodes are more sensitive to temperature changes than silicon or gallium arsenide diodes.

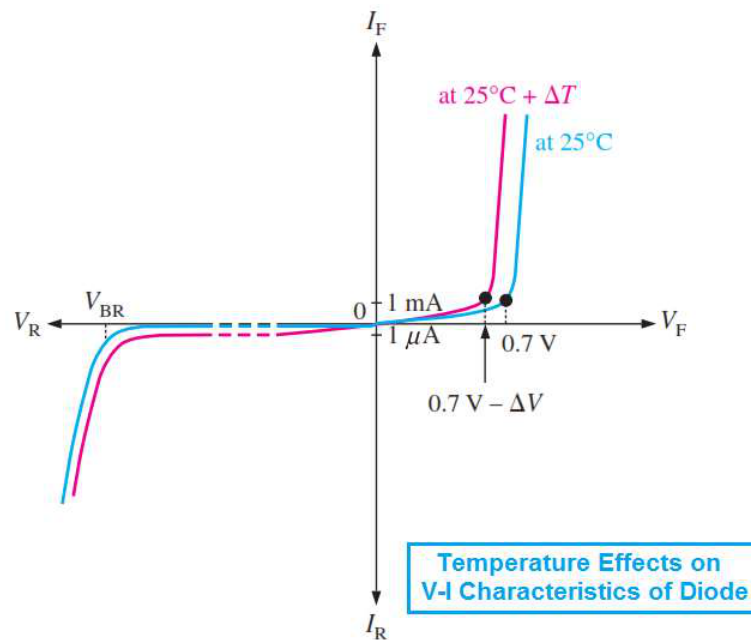


Figure 3: Effect of temperature on forward and reverse characteristics.

Q5. Draw the circuit of a half-wave rectifier and explain its working. State its performance parameters.

A half-wave rectifier uses a single diode in series with the load. The transformer steps the AC mains down to a usable level. During the positive half-cycle the diode is forward biased and conducts, so the positive half appears across the load. During the negative half-cycle the diode is reverse biased and blocks, so the output is zero. A filter capacitor across the load stores energy at the peaks and releases it during troughs, smoothing the output.

Performance: ripple factor = 1.21 (poor DC quality); maximum efficiency = 40.6%; output frequency = input frequency f_{in} ; PIV of the diode = V_m (peak input voltage).

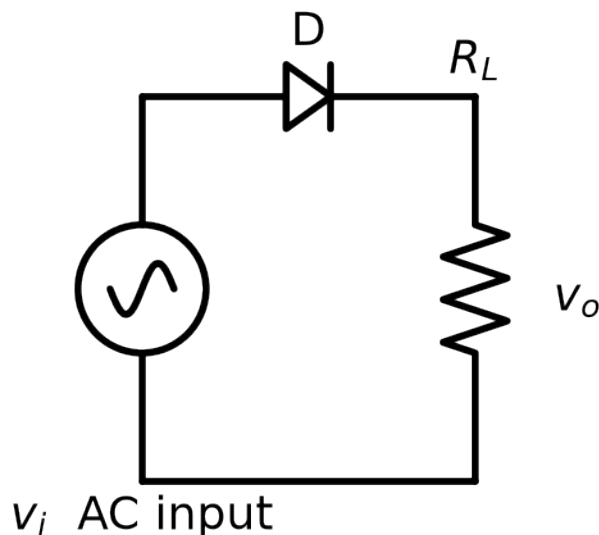


Figure 4: Half-wave rectifier circuit.

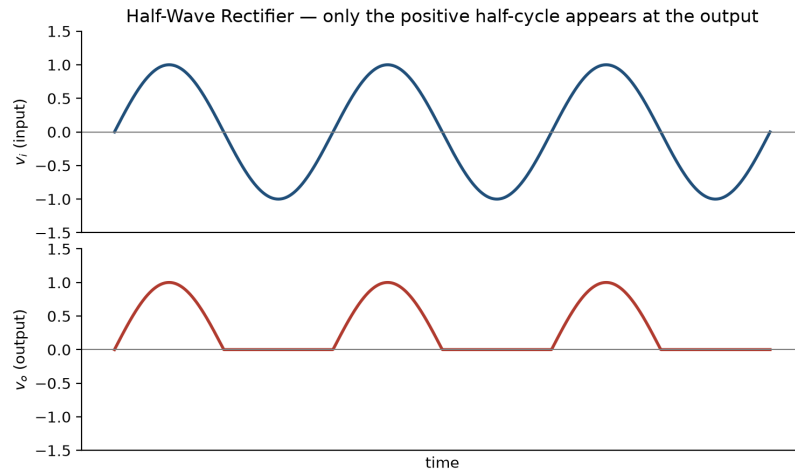


Figure 5: Input and output waveforms of the half-wave rectifier.

Q6. Draw the circuit of a bridge full-wave rectifier and explain its working.

A bridge rectifier uses four diodes connected in a bridge. During the positive half-cycle, D_1 and D_4 (say) conduct and current flows through the load in one direction; during the negative half-cycle, D_2 and D_3 conduct and current flows through the load in the same direction. Thus both half-cycles are delivered to the load.

- Ripple factor = 0.48; maximum efficiency = 81.2%.
- Output frequency = $2 f_{in}$; PIV of each diode = V_m .
- A standard (non-centre-tapped) transformer is sufficient, so it is cheaper and more common than the centre-tapped type.

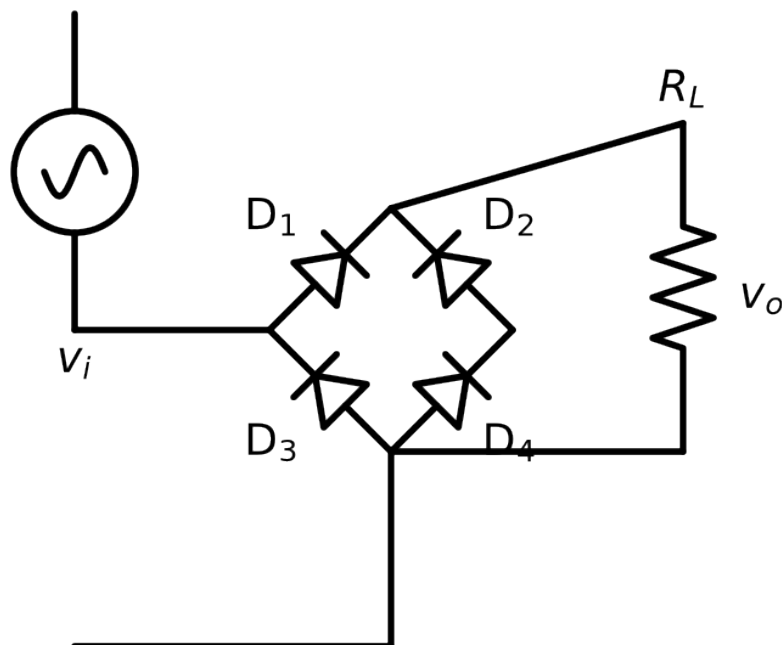


Figure 6: Bridge full-wave rectifier circuit.

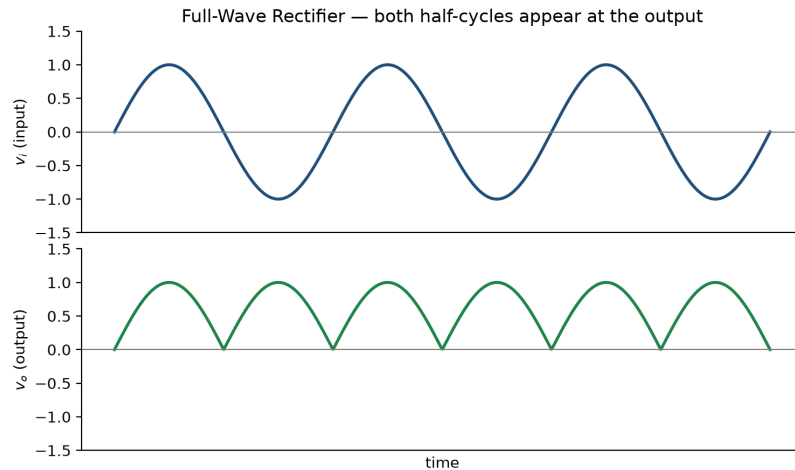


Figure 7: Both half-cycles appear at the output.

Q7. Explain how a Zener diode works as a voltage regulator.

The Zener diode is connected in parallel with the load, in reverse bias, with a series resistor R_S . Because the Zener is heavily doped, it breaks down at a low, well-defined voltage V_Z . Once in breakdown, the voltage across it (and hence across the load) stays almost constant at V_Z even when the input voltage or load current changes.

- 1 Input V_{in} (greater than V_Z) is applied through R_S .
- 2 The output equals V_Z : $V_{out} = V_Z$.
- 3 If V_{in} rises, the extra drop appears across R_S ; the Zener simply passes more current.
- 4 If the load draws more current, the Zener passes less — the total current adjusts, output stays at V_Z .

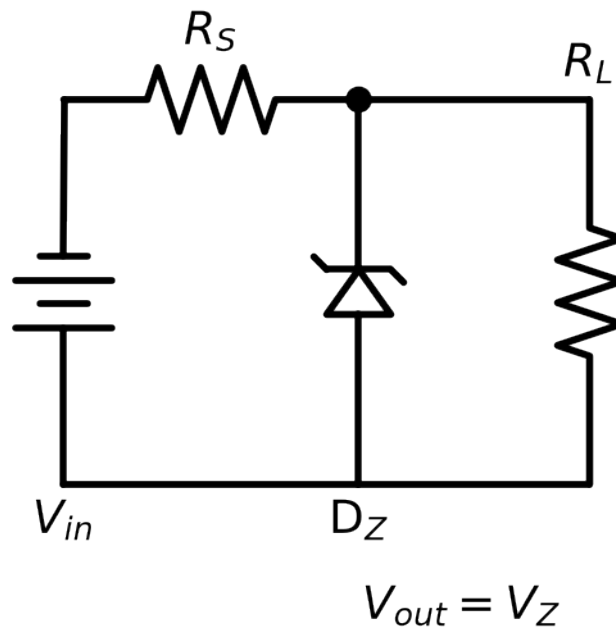


Figure 8: Zener voltage regulator circuit.

Q8. Explain the ideal, simplified and approximate equivalent models of a diode.

The forward voltage V_F applied across an actual diode must overcome (i) the potential barrier V_0 and (ii) the internal drop $I_F r_f$: $V_F = V_0 + I_F r_f$.

- **Ideal model** — the diode is a perfect switch: zero forward drop, infinite reverse resistance. Used for quick estimates.

- **Simplified model** — ideal diode + barrier potential V_0 (internal resistance neglected). Suitable for most analysis.
- **Approximate model** — ideal diode + V_0 + internal dynamic resistance r_f . Most accurate of the three.

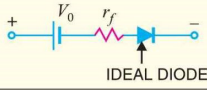
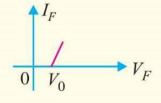
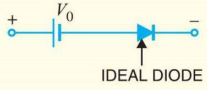
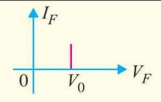
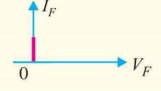
S.No.	Type	Model	Characteristic
1.	Approximate model		
2.	Simplified model		
3.	Ideal Model		

Figure 9: The three diode models with their I–V characteristics.

Q9. Compare half-wave and full-wave rectifiers (any five points).

Parameter	Half-Wave	Full-Wave (Bridge / Centre-Tap)
Number of diodes	1	4 / 2
Ripple factor	1.21	0.48
Maximum efficiency	40.6%	81.2%
Output frequency	f_{in}	$2 f_{in}$
PIV	V_m	V_m (bridge) / $2V_m$ (centre-tap)
Transformer	Standard	Standard (bridge) / centre-tapped

Section C — 7-Mark Questions

Q1. Explain the construction and V–I characteristics of a PN junction diode in detail.

Construction: a PN junction diode is formed by joining a p-type and an n-type semiconductor. It is the fundamental building block of diodes, transistors and integrated circuits. It has three operating conditions — no bias, forward bias and reverse bias.

No bias: electrons and holes diffuse across the junction and recombine, forming a depletion region of immobile ions. The resulting electric field creates a barrier potential (0.7 V for Si, 0.3 V for Ge), and the net current is zero.

Forward bias (p-side to positive): the barrier is reduced, carriers cross the junction, and beyond the knee voltage (about 0.7 V for Si, 0.3 V for Ge) the current increases exponentially.

Reverse bias (n-side to positive): the barrier increases and the depletion region widens; only the small reverse saturation current I_s flows. Beyond the breakdown voltage the current rises sharply and can damage the diode.

The entire behaviour is described by Shockley's equation $I_D = I_s(e^{V_D/nV_T} - 1)$.

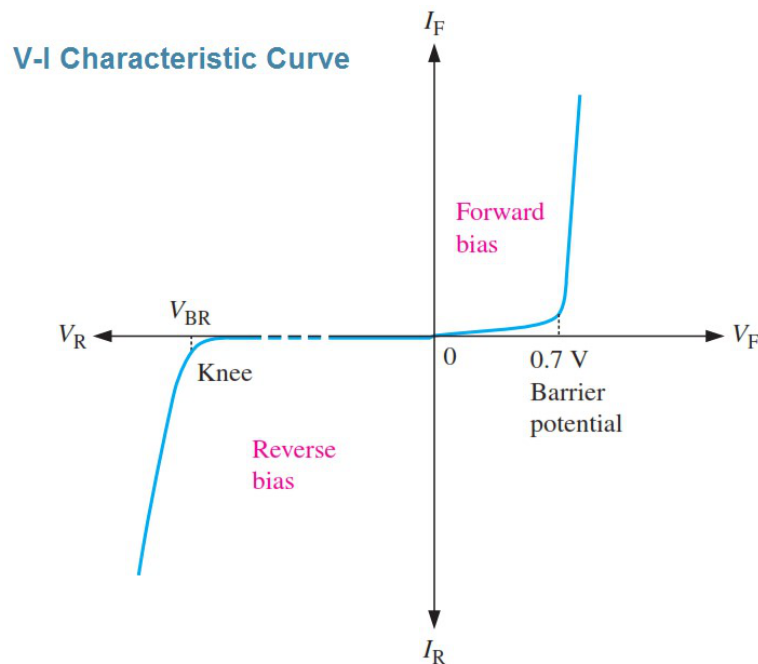


Figure 10: The complete V–I characteristics with forward, reverse and breakdown regions.

Q2. Draw the circuits of a half-wave rectifier and a full-wave rectifier. Explain their working and compare their performance parameters.

Half-wave rectifier: a single diode passes only the positive half-cycle to the load; the negative half is blocked. A transformer steps the mains down, and a filter capacitor smooths the pulsating DC. Ripple factor 1.21, maximum efficiency 40.6%, output frequency f_{in} , $PIV = V_m$.

Full-wave rectifier: uses both half-cycles. In the centre-tapped type, two diodes feed the load in alternate half-cycles and the load returns to the centre tap ($PIV = 2V_m$, needs a special transformer). In the bridge type, four diodes steer both half-cycles through the load in the same direction ($PIV = V_m$, standard transformer). Ripple factor 0.48, maximum efficiency 81.2%, output frequency $2f_{in}$.

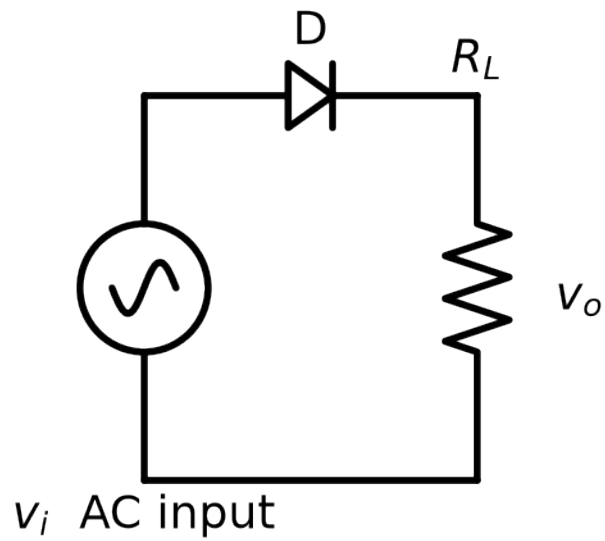


Figure 11: Half-wave rectifier circuit.

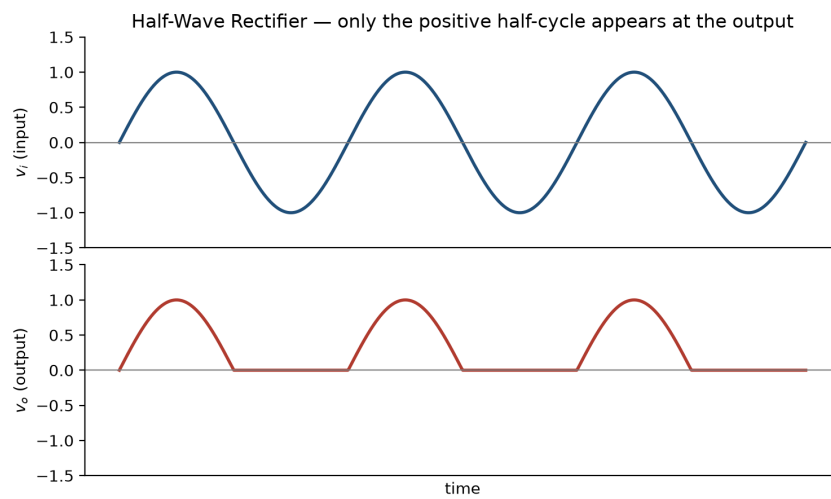


Figure 12: Half-wave rectifier waveforms.

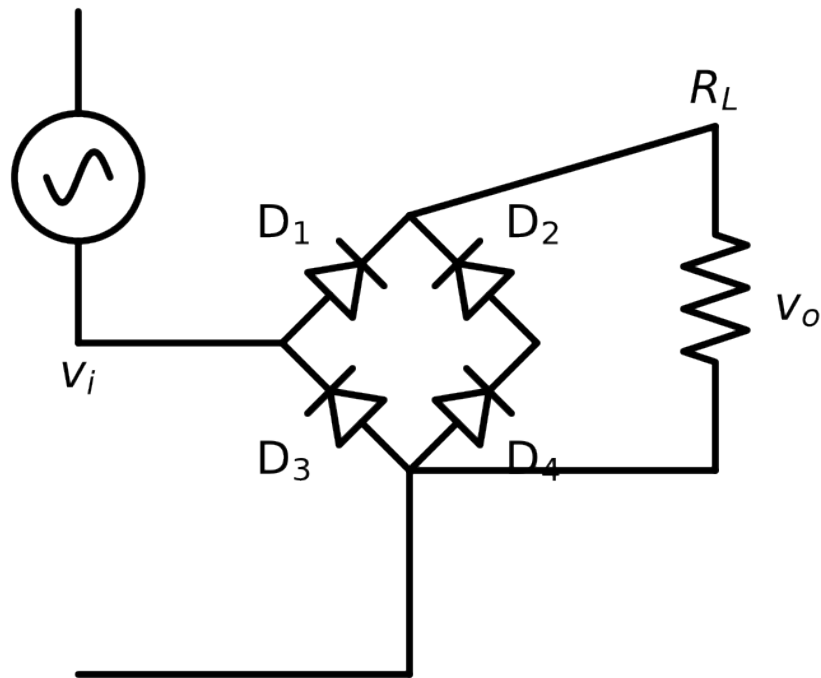


Figure 13: Bridge full-wave rectifier circuit.

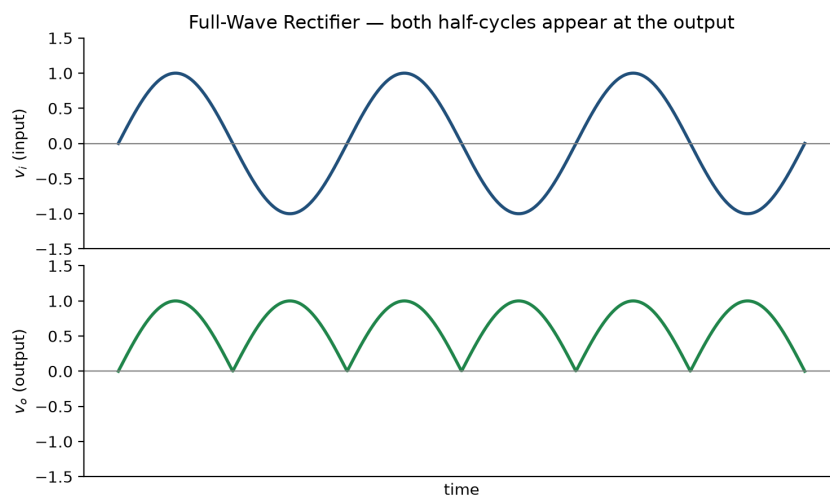


Figure 14: Full-wave rectifier waveforms.

Q3. Explain diode clipper circuits (series and shunt) with circuits, waveforms and applications.

A **clipper (limiter)** removes a portion of the input waveform — positive half, negative half, or both — without distorting the remaining part. It is used for waveform shaping and protection of circuits.

Series clipper: the diode is in series with the load. When the diode is OFF no current flows and the output is zero (clipped); when it is ON the input passes to the output. Reversing the diode selects which half is clipped.

Shunt (parallel) clipper: the diode is in parallel with the load. When the diode is ON it short-circuits the output (clipping it); when OFF the signal passes to the load.

Applications: protection of circuits from overvoltage, removing noise peaks, waveform shaping, and generating square-ish waveforms from sine waves.

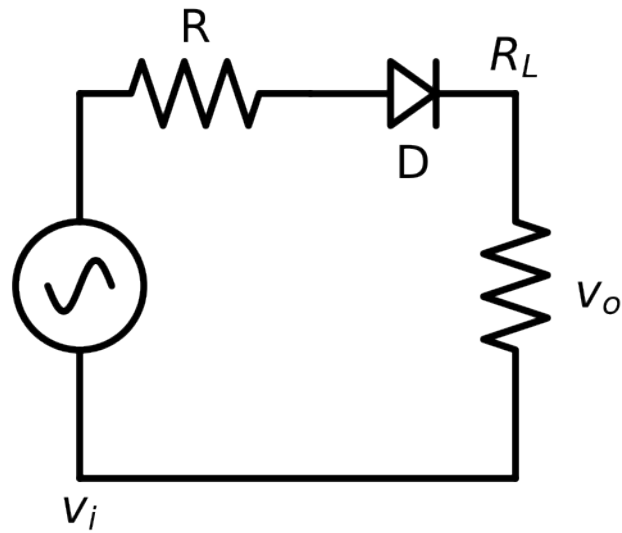


Figure 15: Series clipper.

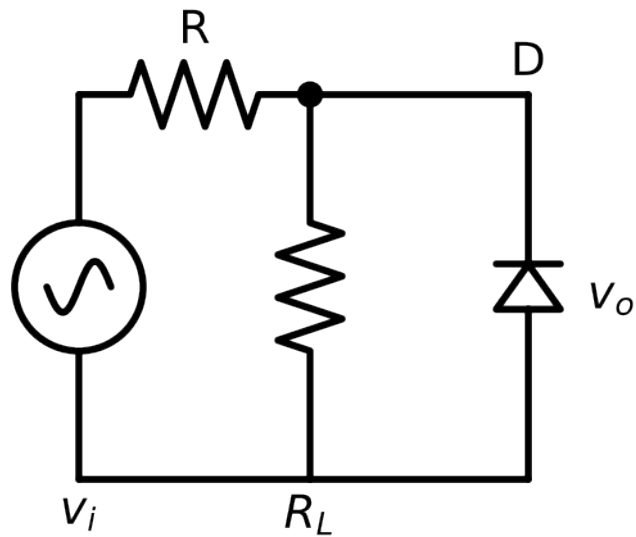


Figure 16: Shunt (parallel) clipper.

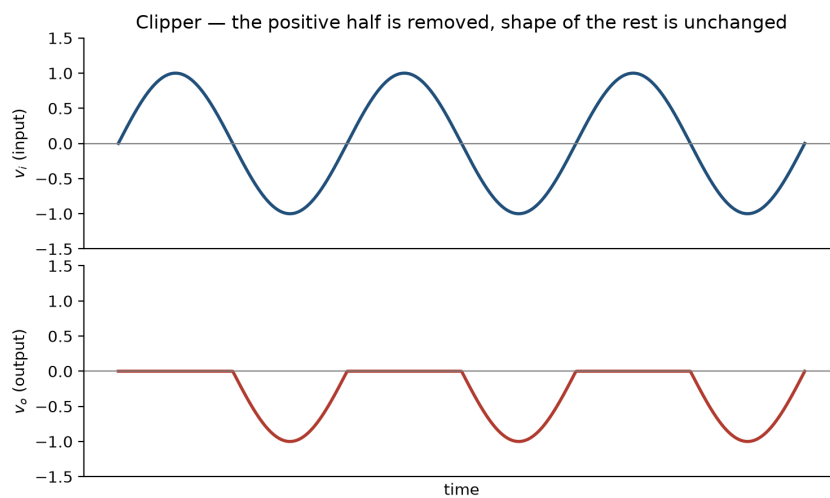


Figure 17: Clipping action on the waveform.

Q4. Explain positive and negative diode clampers with circuits and waveforms.

A **clammer (DC restorer)** shifts the entire waveform up or down by adding a DC component to the signal. It does not change the shape of the waveform — only its DC level (baseline). A capacitor is essential for clamper operation.

Positive clamper: shifts the waveform upward so that the negative peak is clamped to zero (or a reference level) and the whole waveform sits above the x-axis. During the negative half-cycle the diode conducts and charges the capacitor; the charged capacitor then adds its voltage to the input.

Negative clamper: the diode is reversed; the waveform shifts downward, the positive peak is clamped to zero, and the entire waveform sits below the x-axis.

Applications: restoring the DC level of a signal after AC coupling, DC restoration in TV receivers, and level shifting in signal processing.

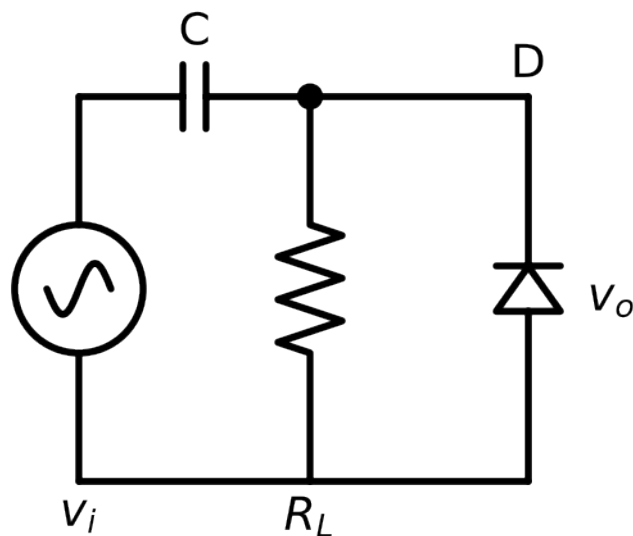


Figure 18: Positive clamper circuit.

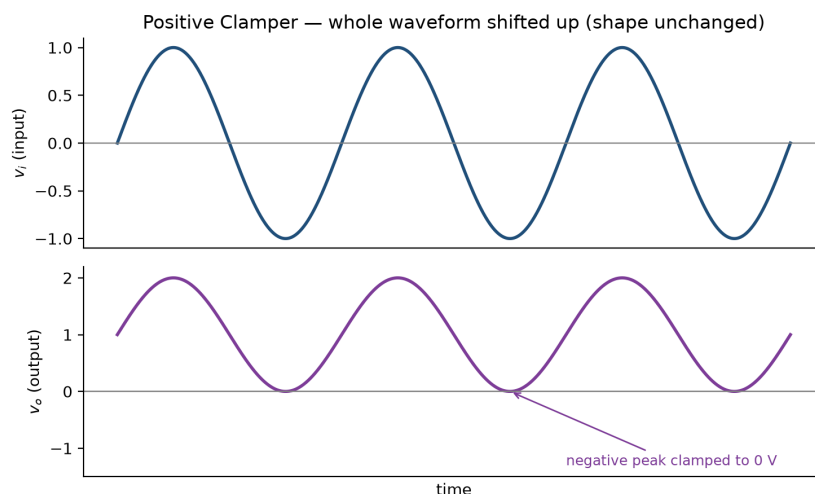


Figure 19: Positive clamper — waveform shifted up, shape unchanged.

Q5. Explain Zener breakdown and avalanche breakdown. Compare them and state the applications of Zener diodes.

Avalanche breakdown occurs in lightly doped diodes where the depletion region is wide. A very large electric field at the junction ruptures covalent bonds, generating a large number of carriers — this multiplication is called avalanche multiplication. It has a positive temperature coefficient (more voltage needed at higher temperature).

Zener breakdown occurs in heavily doped diodes where the depletion region is narrow. Even a small reverse voltage ruptures covalent bonding and generates a large number of carriers. It has a negative temperature coefficient (less voltage needed at higher temperature).

Property	Zener Breakdown	Avalanche Breakdown
Doping level	Heavily doped	Lightly doped
Depletion width	Narrow	Wide
Voltage required	Low (typically < 5 V)	High
Temperature coefficient	Negative	Positive

Applications of Zener diodes: voltage regulation, overvoltage protection, reference voltage sources, wave shaping and meter protection.

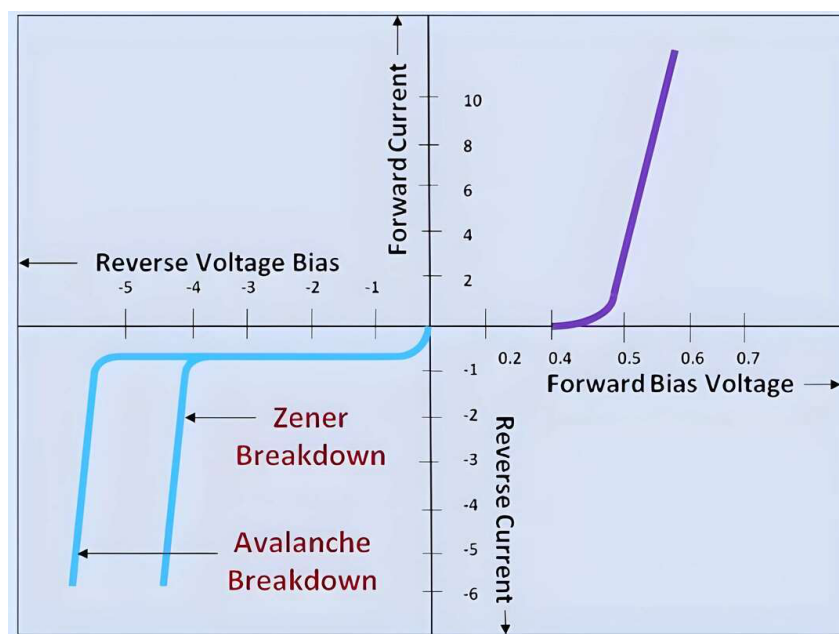


Figure 20: Zener diode I–V characteristic with the breakdown region.

Q6. A germanium diode has a reverse saturation current of 1 μ A at 300 K. Calculate the static and dynamic resistances when the diode is forward biased with 0.25 V ($n = 1$).

Given: $I_0 = 1 \mu\text{A}$, $V_D = 0.25 \text{ V}$, $T = 300 \text{ K}$, $n = 1$.

$$\text{Thermal voltage: } V_T = T_K / 11600 = 300 / 11600 = 25.862 \text{ mV}$$

$$\text{Diode current (Shockley): } I_D = 1 \mu\text{A} \times (e^{0.25 / (1 \times 25.862 \text{ mV})} - 1) = 15.782 \text{ mA}$$

$$\text{Static resistance: } R_D = V_D / I_D = 0.25 \text{ V} / 15.782 \text{ mA} = 15.84 \text{ } \blacksquare$$

$$\text{Dynamic resistance: } r_d = nV_T / I_D = 25.862 \text{ mV} / 15.782 \text{ mA} = 1.638 \text{ } \blacksquare$$

(This numerical pattern appears frequently: find I_D from Shockley's equation first, then apply the two resistance definitions.)